ASSIGNMENT 02 – PFP191

Duration: 70'

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Software Requirements

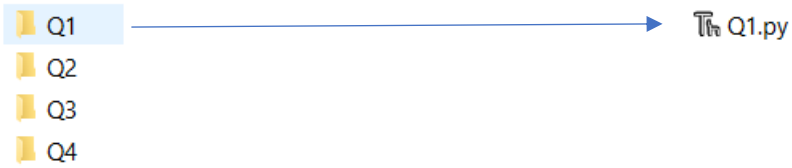
- Thonny, Notepad, Command Prompt, WinRAR / WinZip (or compress to ZIP file of Windows), Windows Explorer (File Explorer) on Windows 7 and above.

Students are ONLY Allowed to use:

- The materials like sample codes and program examples are stored on computer only.

Instructions

- Step 1: Students download the given materials from PEA Client.
- Step 2: Students read questions and prepare answers in the given template.
- Step 3: Submit a solution for each question:
  - The result is one folder for each question (Q1,Q2,Q3,Q4…), which contains **.py** source files.
    Example: Folder "Q1" contains Q1.py for question 1



  - For each question, you must submit one folder as above to PEA Client.

❖ **Importance:**

- Solutions will be marked by Automated Marking Software.
- **The use of tools other than those allowed in the above section whether intentionally or unintentionally, is considered a violation of the exam rules, and the mark is 0**
- Do not change the names of the folders, files, and, files and structure (format) of .py files specified in the exam. If you change it, the marking software can not find the execute file (.py) or the program output to mark, thus the score will be 0
- Do not edit given statements in the **main** function. If you change, the marking software can not mark and the score will be 0.

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Question 1: (3 marks)

The given file Q1.py contains statements to input data for 3 integers a, b and c, which are the scores of subjects of a student. You should write statements to calculate the avg average of the above subjects and print the result, which can be classified as "Excellent" if avg >=9, "Good" if avg >=8, "Average" if avg> =7, and "Poor" otherwise.

Notes:

- The output result is formatted in two decimal places

Sample input and output:

Input: a = 9, b = 9 and c = 9 After processing: 9.00 and Excellent Output for marking: OUTPUT: 9.00 Excellent	Input: a = 8, b = 8 and c = 8 After processing: 8.00 and Good Output for marking: OUTPUT: 9.00 Good
Input: a = 6, b = 6 and c = 6 After processing: 9.00 and Average Output for marking: OUTPUT: 9.00 Average	Input: a = 4, b = 4 and c = 5 After processing: 4.33 and Poor Output for marking: OUTPUT: 9.00 Poor

Question 2:

(2 marks)

The given file Q2.py contains the commands to input the temperature and the temperature selection (Celsius, Fahrenheit and Kelvin). You should write the commands to convert the three temperatures.

Hint:

- Fahrenheit = (Celsius * 9.0 / 5.0) + 32;
- Kelvin = Celsius + 273.15;

Notes:

- The output result is formatted in two decimal places

Sample input and output:

Input: 37 and 1 After processing: 37.00, 98.60, 310.15 Output for marking: OUTPUT: 37.00 98.60 310.15	Input: 98.60 and 2 After processing: 37.00, 98.60, 310.15 Output for marking: OUTPUT: 37.00 98.60 310.15	Input: 310.15 and 3 After processing: 37.00, 98.60, 310.15 Output for marking: OUTPUT: 37.00 98.60 310.15

Question 3:

(2 marks)

The given file Q3.py contains statements to input data for two integers a, b. You should write statements to print value of T.

$$T = \begin{cases} a^2 + ab + \sqrt{ab} & \mathbf{a < b} \quad AND \quad \mathbf{a < 10} \\ ab^2 - 2a + 5b & \mathbf{a = b} \quad OR \quad \mathbf{b > 10} \end{cases}$$

Notes:

- The output result is formatted in two decimal places

Sample input and output:

Input: a = 5, b=9 After processing: 76.71 Output for marking: OUTPUT: 76.71	Input: a = 9, b=9 After processing: 756.00 Output for marking: OUTPUT: 756.00	Input: a = 9, b= 8 After processing: 0 Output for marking: OUTPUT: 0.00

Question 4:

(3 marks)

The given file Q4.py contains statements to input data for non-negative integers less than 100. You should write statements to print that number in words.

Sample input and output:

Input: 80 After processing: Eight Zero Output for marking: OUTPUT: Eight Zero	Input: 8 After processing: Eight Output for marking: OUTPUT: Eight	Input: 18 After processing: One Eight Output for marking: OUTPUT: One Eight